CHAPTER 04

Analyzing Organic Reactions

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

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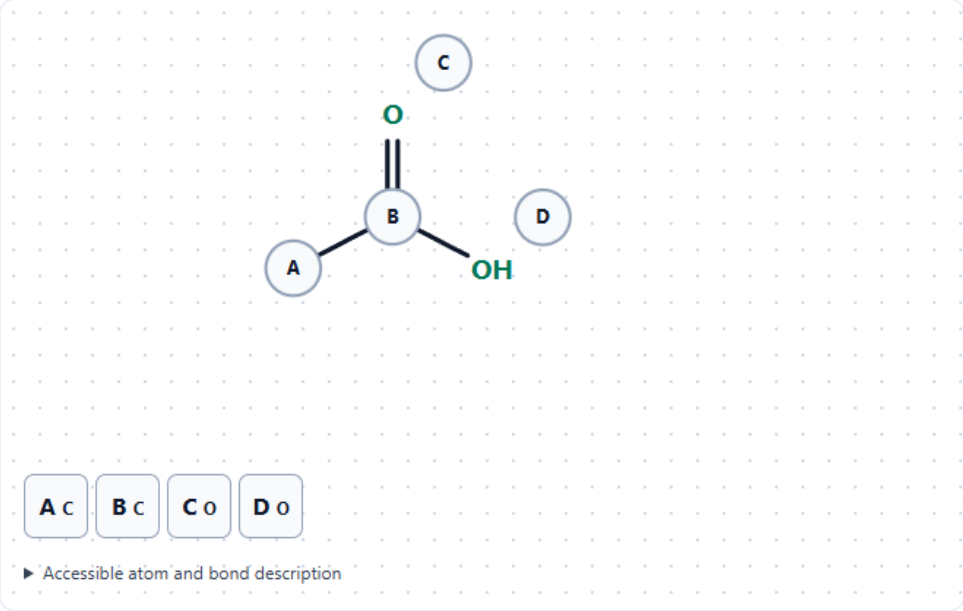
BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

pka-equilibrium | Independent exercise

The displayed carboxyl group is a structural example for Stage 1. Stages 2–3 use a separate hypothetical proton-transfer equilibrium, HA + B ⇌ A− + BH+, with all pKa values on a comparable solvent/temperature scale; they are not measured values for the displayed ethanoic acid.

Screenshot: pka-equilibrium-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

pka-equilibrium:stage-1 | Select both oxygens of the carboxyl group.

Answer type: atoms

pka-equilibrium:stage-2 | For HA + B ⇌ A− + BH+, if pKa(HA)=5 and pKa(BH+)=10, equilibrium favors…

Answer type: choice

Option 0 (A): Reactants

Option 1 (B): Products

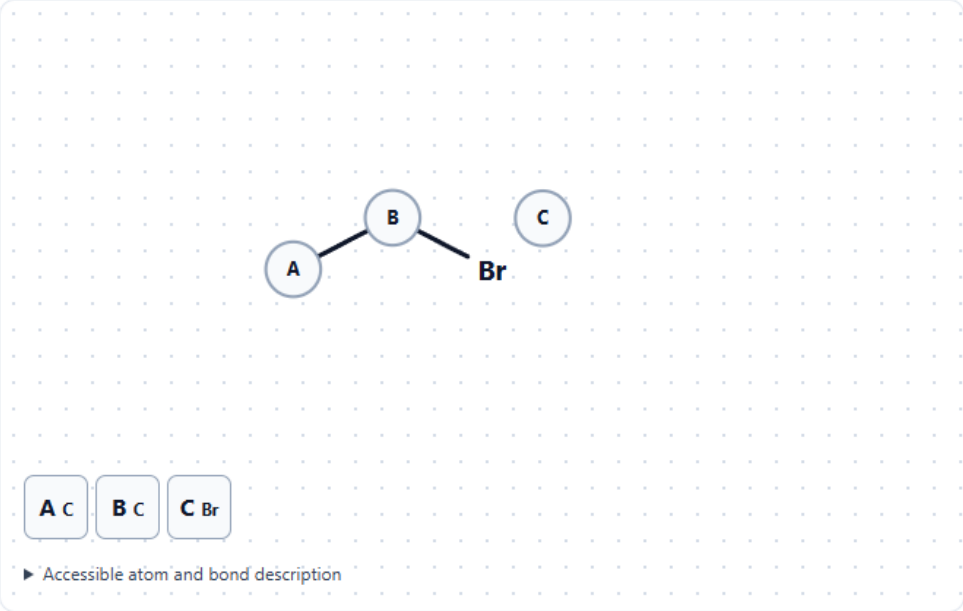
Option 2 (C): Exactly equal amounts

pka-equilibrium:stage-3 | For that forward equilibrium, using pKa(HA)=5 and pKa(BH+)=10, estimate K.

Answer type: number | unit: dimensionless

identify-partners | Independent exercise

Screenshot: identify-partners-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): Br; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

identify-partners:stage-1 | Select the electrophilic carbon.

Answer type: atoms

identify-partners:stage-2 | Among these choices (including a reagent not drawn), which species supplies a lone pair as a nucleophile?

Answer type: choice

Option 0 (A): Hydroxide ion

Option 1 (B): A proton

Option 2 (C): The electron-poor carbon

identify-partners:stage-3 | When C–Br breaks heterolytically, the pair goes to…

Answer type: choice

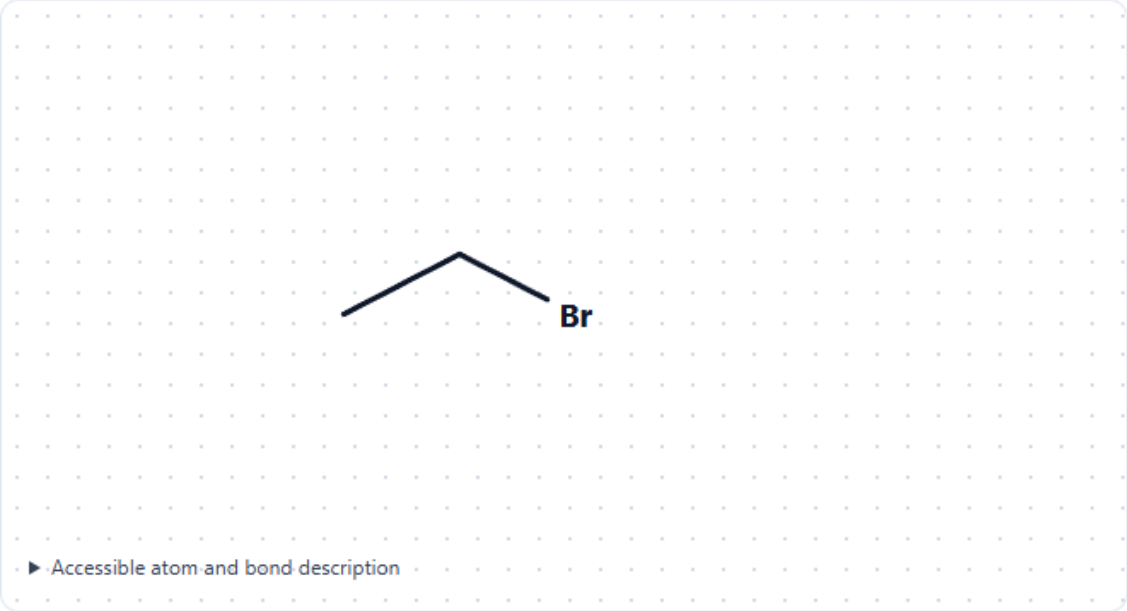
Option 0 (A): Carbon

Option 1 (B): Bromine

Option 2 (C): Neither atom

sn2-rate | Independent exercise

Screenshot: sn2-rate-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): Br; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

sn2-rate:stage-1 | With a small, strong nucleophile in a suitable polar aprotic solvent, this primary halide favors...

Answer type: choice

Option 0 (A): SN2

Option 1 (B): Unstabilized primary SN1

sn2-rate:stage-2 | If both substrate and nucleophile concentrations double, by what factor does the ideal SN2 rate change?

Answer type: number | unit: fold

sn2-rate:stage-3 | Apply the general rule to a separate chiral SN2 center (the displayed bromoethane is achiral): its geometry undergoes...

Answer type: choice

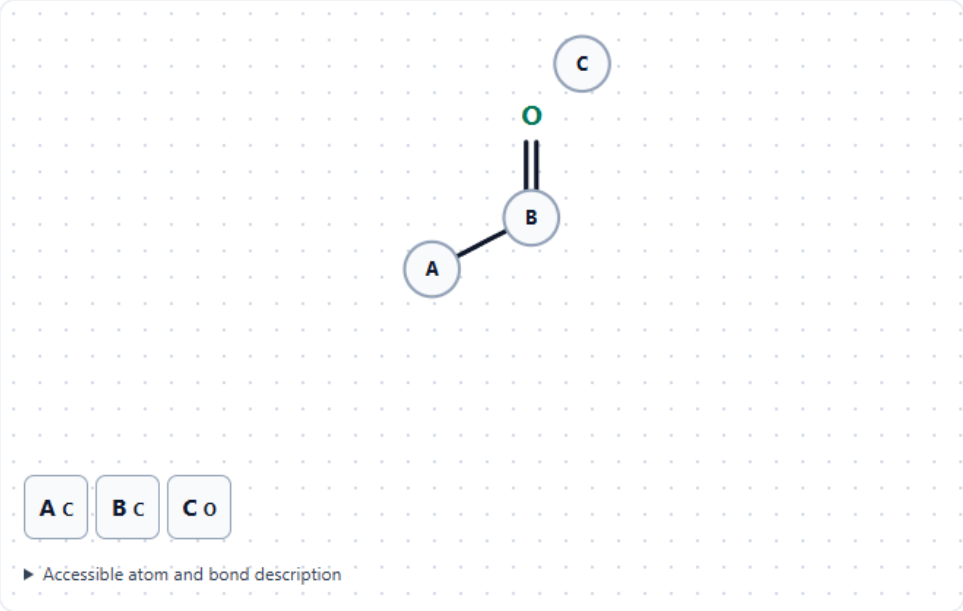
Option 0 (A): Inversion

Option 1 (B): Always complete racemization

Option 2 (C): Retention

carbon-redox | Independent exercise

Screenshot: carbon-redox-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

carbon-redox:stage-1 | Select the carbonyl carbon.

Answer type: atoms

carbon-redox:stage-2 | What is this aldehyde carbon’s oxidation state (a dimensionless signed number)?

Answer type: number | unit: dimensionless

carbon-redox:stage-3 | NaBH4 followed by protonation converts it to…

Answer type: choice

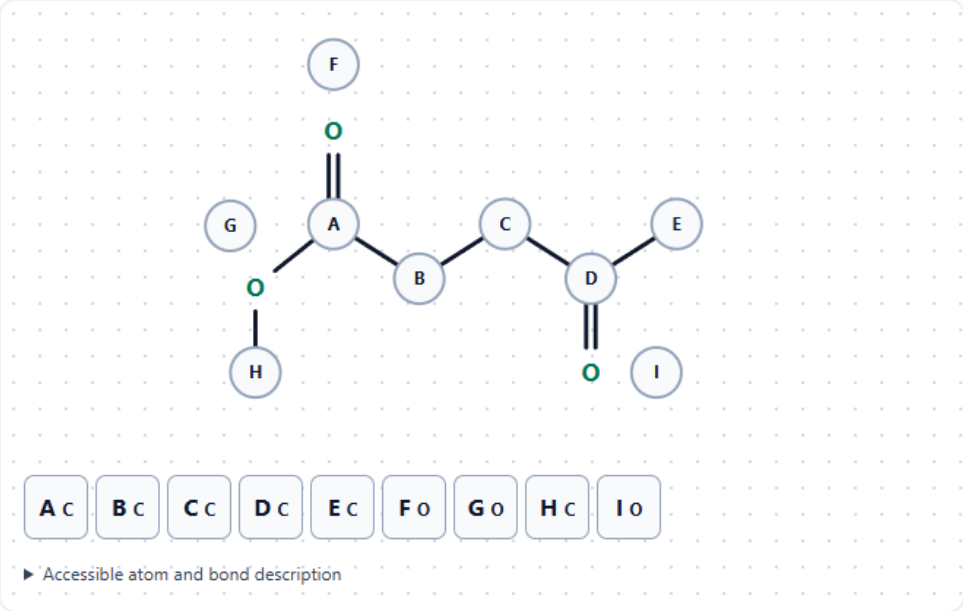
Option 0 (A): A primary alcohol

Option 1 (B): A carboxylic acid

Option 2 (C): An alkane under ordinary conditions

selective-reduction | Independent exercise

Screenshot: selective-reduction-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=70.00, y=100.00; charge=0; explicit H override=none

1 (B): C; x=125.00, y=135.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=235.00, y=135.00; charge=0; explicit H override=none

4 (E): C; x=290.00, y=100.00; charge=0; explicit H override=none

5 (F): O; x=70.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=20.00, y=140.00; charge=0; explicit H override=none

7 (H): C; x=20.00, y=195.00; charge=0; explicit H override=none

8 (I): O; x=235.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [0, 5, 2], [0, 6, 1], [6, 7, 1], [3, 8, 2]]

selective-reduction:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

selective-reduction:stage-2 | Select the ketone carbonyl carbon preferred by ordinary NaBH4 conditions.

Answer type: atoms

selective-reduction:stage-3 | Why not simply select the highest-priority naming group?

Answer type: choice

Option 0 (A): Naming seniority does not determine kinetic reactivity

Option 1 (B): The ester has no carbonyl

Option 2 (C): All carbonyls react identically